

# Rational Descent and the Rank-Ten Dihedral Core of the Fourth Hénon Moment Packet

Hilbert–Pólya Dynamical Structure Exploration Project

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## Abstract

Let  $K = \mathbf{Q}(\rho)$ , with  $\rho^2 + \rho + 1 = 0$ , and let

$$X_n = \left\{ \sum_{i=0}^{2n-1} x_i^3 = \sum_{i=0}^{2n-2} x_i x_{i+1} + \rho x_{2n-1} x_0 = 0 \right\} \subset \mathbf{P}_K^{2n-1}.$$

These source-ordered complete intersections encode the cohomological moments of a normalized Hénon Euler germ. We give an explicit semilinear reversal and a closed fixed basis that descend  $X_n$  to  $\mathbf{Q}$  for every  $n \geq 2$ . The equation theorem is unconditional; the motivic statements use only the certified smooth rows  $n = 2, 3, 4$ . Their rational moment packets have ranks 15, 63, 255. At every good split rational prime, the C51 exponent  $2/n$  over the two  $K$ -places becomes  $4/n$  on one rational factor. The fourth-row half-power therefore equals one ordinary rank-255, exponent-one rational local factor, whereas the third row remains fractional with exponent  $4/3$ . For  $n = 4$ , Galois transport twists the order-24 dihedral source group rather than fixing its elements. Its Reynolds graph sum nevertheless descends and splits the rational middle motive into ranks  $10 + 158$ . The raw rank-10 summand has an  $\ell$ -independent degree-10 polynomial in  $\mathbf{Z}[T]$ , pure of weight 5, at every good prime. The split identity does not extend to inert primes and yields no global square root, continuation, or functional equation.

## 1 Introduction

The fourth cohomological moment in the Hénon source program carries an apparently awkward half-power over the quadratic field  $K = \mathbf{Q}(\rho)$ . A direct  $K$ -compatible system cannot realize that half-power without acquiring a nonintegral rank. The obstruction leaves one algebraic escape: the two  $K$ -places above a split rational prime could be the conjugate copies of a single object over  $\mathbf{Q}$ .

This paper constructs that rational object. The construction begins at the equation level, before any local factor is manipulated. Reversal of the source chronology is accompanied by alternating powers of  $\rho$ ; the resulting monomial map is a Weil descent datum. Solving its fixed-vector equations gives one rational coordinate system in every row and preserves the weighted closing edge. Thus the repair is geometric rather than a formal choice of a square-root branch.

The fourth row contains more structure. The preceding source analysis found an order-24 group  $\text{Dih}(C_{12})$  and a rank-10 Reynolds summand in the middle cohomology. Conjugation by the rational descent does not fix the twenty-four transformations individually. Instead it acts by

$$r \longmapsto r^{-1}, \quad s \longmapsto sr^{-1}.$$

The correct rational object is consequently a nonconstant finite étale group scheme split by  $K$ . Its graph sum is invariant, so rational restriction and corestriction descend the Chow projector.

## Contributions

The paper proves four claims.

1. For every  $n \geq 2$ , we construct an explicit  $\mathbf{Q}$ -model of the source-ordered  $(2, 3)$  intersection and prove a closed determinant formula for the coordinate change. This statement concerns equations; smoothness and motives are not promoted beyond the certified rows.
2. For  $n = 2, 3, 4$ , we descend the full cohomological packet of rank  $4^n - 1$ . At good split primes, its exponent changes from  $2/n$  over  $K$  to  $4/n$  over  $\mathbf{Q}$ . The  $n = 4$  half-power clears exactly.
3. We descend the order-24 Reynolds correspondence and obtain rational Chow summands of ranks 10 and 158. The rank-10 summand has Hodge ledger  $(1, 4, 4, 1)$  and becomes of Calabi–Yau-threefold Hodge type after one Tate twist.
4. At good primes, the raw rank-10 Frobenius polynomial is  $\ell$ -independent, belongs to  $\mathbf{Z}[T]$ , has weight 5, and obeys the expected degree-five reciprocity. This strict compatibility statement does not use semisimplicity.

## Relation to existing geometry

Favero–Iliev–Katzarkov give the Hodge numbers and Cayley-ring description for smooth  $(2, 3)$  fivefolds in  $\mathbf{P}^7$  [2, §5.1, equation (6), and §5.4, equation (8)]. Those results supply context for the ledger, not the source-specific descent or projector. Laterveer records ordinary nonmiddle Chow–Künneth projectors for intersections of a quadric and a cubic [6, proof of Theorem 4.1]; the multiplicative theorem there assumes even ambient dimension and does not apply to  $\mathbf{P}^7$ . Vial gives broader finite-dimensional-motive context [8, Example 4.12]. None of these references identifies the Hénnon-derived rational form or its twisted dihedral core.

Our descent proof is explicit and therefore avoids a general effectivity claim for arbitrary projective descent data. For Chow cycles, we use the standard proper-pushforward and flat-pullback degree formulas [3, §§1.4, 1.7]. Kahn’s motivic descent theorem [4, Theorems 1 and 7.1] gives a modern categorical backdrop, but the graph-sum argument below is elementary with rational coefficients. For local polynomials, the comparison input is the correspondence-trace theorem of Katz–Messing [5, Theorem 2(2)].

## Scope

Every Frobenius in this paper is geometric, normalized by  $\text{Frob}_p \mid \mathbf{Q}_\ell(-1) = p$ . The rational exponent clearing is local to good split primes. At inert primes, Frobenius eigenvalues are squared rather than duplicated, which blocks a global half-root. We do not claim smoothness for  $n \geq 5$ , automorphy, meromorphic continuation, a functional equation, a Calabi–Yau realization, or Chow indecomposability. The exact irreducibility gate for the rank-10 polynomial is recorded only as a successor problem.

## 2 The source family and main theorem

Let

$$K = \mathbf{Q}(\rho), \quad \rho^2 + \rho + 1 = 0, \quad \tau(\rho) = \rho^2.$$

For  $n \geq 2$ , set  $N = 2n$  and

$$C_n = \sum_{i=0}^{N-1} x_i^3, \quad Q_{n,\rho} = \sum_{i=0}^{N-2} x_i x_{i+1} + \rho x_{N-1} x_0. \tag{1}$$

The source variety is

$$X_n = V(C_n, Q_{n,\rho}) \subset \mathbf{P}_K^{N-1}.$$

The order of the variables and the coefficient on  $x_{N-1}x_0$  are part of the definition.

Define  $\sigma(i) = -i \bmod N$ , and put

$$e_i = \begin{cases} 1, & i \neq 0 \text{ and } i \text{ is even,} \\ 0, & i = 0 \text{ or } i \text{ is odd.} \end{cases}$$

Let  $M_n$  be the monomial transformation

$$(M_n x)_i = \rho^{e_i} x_{\sigma(i)}. \quad (2)$$

Finally set  $\theta = 1 + 2\rho$ , so that  $\theta^2 = -3$  and  $\tau(\theta) = -\theta$ .

**Theorem 2.1** (Explicit family descent). *For every  $n \geq 2$ ,*

$$C_n(M_n x) = C_n(x), \quad Q_{n,\rho}(M_n x) = \rho Q_{n,\rho^2}(x), \quad M_n \tau(M_n) = I.$$

*In rational coordinates*

$$(u_0, a_1, b_1, \dots, a_{n-1}, b_{n-1}, c),$$

*define  $B_n$  by*

$$\begin{aligned} x_0 &= u_0, \\ x_i &= a_i + \theta b_i, \quad x_{N-i} = \rho^{e_i}(a_i - \theta b_i) \quad (1 \leq i < n), \\ x_n &= \kappa_n c, \quad \kappa_n = \begin{cases} 1, & n \text{ odd,} \\ 1 + \rho, & n \text{ even.} \end{cases} \end{aligned} \quad (3)$$

*Then*

$$M_n \tau(B_n) = B_n, \quad \det B_n = (2\theta)^{n-1} \rho^{\lfloor (n-1)/2 \rfloor} \kappa_n \neq 0. \quad (4)$$

*The forms*

$$C_{n,0} = u_0^3 + \sum_{i=1}^{n-1} (2a_i^3 - 18a_i b_i^2) + (-1)^{n+1} c^3 \quad (5)$$

*and*

$$\begin{aligned} Q_{n,0} &= u_0(a_1 + 3b_1) \\ &\quad + \sum_{i=1}^{n-2} (a_i a_{i+1} + 3a_i b_{i+1} + 3b_i a_{i+1} - 3b_i b_{i+1}) + R_n, \\ R_n &= \begin{cases} (a_{n-1} + 3b_{n-1})c, & n \text{ odd,} \\ 2a_{n-1}c, & n \text{ even} \end{cases} \end{aligned} \quad (6)$$

*satisfy*

$$C_n(B_n u) = C_{n,0}(u), \quad Q_{n,\rho}(B_n u) = (1 + \rho)Q_{n,0}(u).$$

*Consequently*

$$X_{n,0} = V(C_{n,0}, Q_{n,0}) \subset \mathbf{P}_{\mathbf{Q}}^{2n-1}$$

*base-changes to  $X_n$ .*

Theorem 2.1 is an identity theorem for equations. The source work certifies smoothness only for  $n = 2, 3, 4$ ; no later row is declared smooth here.

**Theorem 2.2** (Rational packets and fourth-row repair). *For  $n = 2, 3, 4$ , the normalized even and odd packets descend to a rational packet*

$$W_n = E_n \oplus O_n, \quad \text{rank } W_n = 4^n - 1.$$

*Their ranks are 15, 63, 255. At every good split rational prime,*

$$\frac{2}{n} \text{Log}_0 L_{K,p}(W_{n,K}, u) = \frac{4}{n} \text{Log}_0 L_{\mathbf{Q},p}(W_n, u).$$

*The reduced rational denominator is  $n/\gcd(n, 4)$ . In particular,*

$$(L_{K,p}(W_{4,K}, u))_{\text{Log}_0}^{1/2} = L_{\mathbf{Q},p}(W_4, u). \quad (7)$$

*The right side is one rank-255, exponent-one rational local factor.*

**Theorem 2.3** (Rational dihedral core). *For  $n = 4$ , the order-24 Reynolds middle projector descends to  $\mathbf{Q}$ . It gives mutually orthogonal self-transpose Chow projectors  $\pi_{\text{core},0}$  and  $\pi_{\text{lev},0}$  of ranks 10 and 158. The full rational packet decomposes by ranks*

$$87 + 10 + 158 = 255.$$

*The raw rank-10 summand  $M_0 = (X_{4,0}, \pi_{\text{core},0}, 0)$  has Hodge numbers*

$$h^{4,1} = h^{1,4} = 1, \quad h^{3,2} = h^{2,3} = 4.$$

**Theorem 2.4** (Good-prime polynomial). *Outside a finite set, for every  $\ell \neq p$ ,*

$$P_p(T) = \det(1 - \text{Frob}_p T \mid H_\ell(M_0)) = \sum_{k=0}^{10} a_k T^k$$

*is a degree-10 polynomial in  $\mathbf{Z}[T]$ , independent of  $\ell$ , and pure of weight 5. It satisfies*

$$a_{10-k} = p^{25-5k} a_k \quad (0 \leq k \leq 10).$$

*The twice-twisted system  $M_0(2)$  has weight 1 and local polynomial  $P_p(T/p^2) \in \mathbf{Q}[T]$ .*

### 3 The semilinear reversal and its fixed basis

#### 3.1 The cocycle identities

*Proof of the first part of Theorem 2.1.* Because  $\rho^3 = 1$  and  $\sigma$  is a permutation,

$$C_n(M_n x) = \sum_i \rho^{3e_i} x_{\sigma(i)}^3 = C_n(x).$$

For the quadric, separate the edge indexed by  $i = 0$ , the edges  $1 \leq i \leq N - 2$ , and the closing edge. Reversal sends the  $i = 0$  edge to the closing pair with phase 1. Every edge indexed by  $1 \leq i \leq N - 2$  has exactly one nonzero even endpoint and therefore acquires phase  $\rho$ . The closing term in  $Q_{n,\rho}$  is sent to the edge  $x_0 x_1$ , still with coefficient  $\rho$ . Hence

$$Q_{n,\rho}(M_n x) = x_{N-1} x_0 + \rho \sum_{i=0}^{N-2} x_i x_{i+1} = \rho Q_{n,\rho^2}(x).$$

Finally  $e_{\sigma(i)} = e_i$ . Applying the conjugate matrix after  $M_n$  multiplies coordinate  $i$  by

$$\rho^{e_i} \tau(\rho^{e_{\sigma(i)}}) = \rho^{e_i + 2e_i} = 1,$$

which proves  $M_n \tau(M_n) = I$ . □

### 3.2 Fixed vectors

The relation  $\tau(\theta) = -\theta$  makes the pair in (3) fixed:

$$\rho^{e_i} \tau(x_{N-i}) = \rho^{e_i} \rho^{2e_i} (a_i + \theta b_i) = x_i.$$

The reverse identity is the same calculation. At the central index,  $\sigma(n) = n$ . If  $n$  is odd, then  $e_n = 0$  and  $x_n = c$  is fixed. If  $n$  is even, then  $e_n = 1$ , and

$$\rho \tau(1 + \rho) = \rho(1 + \rho^2) = 1 + \rho,$$

so  $x_n = (1 + \rho)c$  is fixed.

**Lemma 3.1** (Determinant). *In the source row order and the rational variable order of Theorem 2.1,*

$$\det B_n = (2\theta)^{n-1} \rho^{\lfloor (n-1)/2 \rfloor} \kappa_n.$$

*Proof.* After pairing rows  $i, N - i$ , the block for  $(a_i, b_i)$  is

$$\begin{pmatrix} 1 & \theta \\ \rho^{e_i} & -\rho^{e_i} \theta \end{pmatrix},$$

with determinant  $-2\rho^{e_i}\theta$ . The permutation from the source row order to the paired order has sign  $(-1)^{n-1}$ , which cancels the product of the  $n - 1$  block signs. Exactly  $\lfloor (n - 1)/2 \rfloor$  indices in  $1, \dots, n - 1$  are even. The unpaired coordinates contribute 1 and  $\kappa_n$ . Multiplying the blocks proves the formula.  $\square$

The formula is nonzero in  $K$ , so fixed vectors supply a full basis. This proves effectivity directly. In particular, no general assertion about arbitrary projective descent data is needed; compare the standard descent cautions and the smoothness-locality statement in Stacks Project Tags 08KE and 02VL [7].

### 3.3 Expansion over the rational field

The cubic calculation uses

$$(a + \theta b)^3 + (a - \theta b)^3 = 2a^3 + 6a\theta^2 b^2 = 2a^3 - 18ab^2.$$

If  $n$  is odd, the central cube is  $c^3$ . If  $n$  is even, then  $(1 + \rho)^3 = -1$ , so the central cube is  $-c^3$ . These two cases give (5).

For adjacent paired coordinates, expansion and  $\theta^2 = -3$  give

$$a_i a_{i+1} + 3a_i b_{i+1} + 3b_i a_{i+1} - 3b_i b_{i+1}$$

after the common scalar  $1 + \rho$  is removed. The initial edge gives  $u_0(a_1 + 3b_1)$ . At the center, the two incident edges give

$$R_n = \begin{cases} (a_{n-1} + 3b_{n-1})c, & n \text{ odd,} \\ 2a_{n-1}c, & n \text{ even.} \end{cases}$$

This proves (6) and completes Theorem 2.1.

### 3.4 The fourth rational model

For  $n = 4$ , write

$$(u_0, u_1, u_2, u_3, u_4, u_5, u_6, u_7) = (u_0, a_1, b_1, a_2, b_2, a_3, b_3, c).$$

The rational model is

$$\begin{aligned} C_{4,0} &= u_0^3 + 2u_1^3 - 18u_1u_2^2 + 2u_3^3 - 18u_3u_4^2 \\ &\quad + 2u_5^3 - 18u_5u_6^2 - u_7^3, \\ Q_{4,0} &= u_0u_1 + 3u_0u_2 + u_1u_3 + 3u_1u_4 + 3u_2u_3 - 3u_2u_4 \\ &\quad + u_3u_5 + 3u_3u_6 + 3u_4u_5 - 3u_4u_6 + 2u_5u_7. \end{aligned}$$

Lemma 3.1 gives  $\det B_4 = 24\theta$ . Smoothness follows from the certified smooth  $K$ -model by base change; this implication does not certify any untested source row.

## 4 Rational packets and the split half-power

Let  $S_n = V(\sum x_i^3)$  over  $\mathbf{Q}$ . For the certified smooth rows  $n = 2, 3, 4$ , define

$$\mathbf{E}_n = \mathbf{1} \oplus (S_n, \pi_{\text{prim}}^{S_n}, n-1), \quad \mathbf{O}_n = (X_{n,0}, \pi_{2n-3}^{X_n}, n-2),$$

and

$$\mathbf{W}_n = \mathbf{E}_n \oplus \mathbf{O}_n.$$

The projectors are the ordinary hyperplane projectors and their middle complements. For the fivefold  $X_{4,0}$ , for example,

$$\pi_{2i} = \frac{1}{6}h^{5-i} \times h^i, \quad \pi_5 = \Delta - \sum_{i=0}^5 \pi_{2i}. \quad (8)$$

The coefficient  $1/6$  is the inverse degree of the complete intersection. No multiplicative Chow–Künneth assertion is used.

The source cohomology calculation gives

$$\text{rank } \mathbf{E}_n = \frac{4^n + 5}{3}, \quad \text{rank } \mathbf{O}_n = \frac{2(4^n - 4)}{3}. \quad (9)$$

Both summands are now defined over  $\mathbf{Q}$ : the Fermat cubic was rational from the outset, and Theorem 2.1 supplies  $X_{n,0}$ . Base change preserves the projectors and ranks. Adding (9) proves

$$\text{rank } \mathbf{W}_n = 4^n - 1,$$

which is 15, 63, 255 in the three certified rows.

### 4.1 Placewise quadratic base change

For a rational prime  $p$  outside the common bad set, write

$$L_{K,p}(\mathbf{W}_{n,K}, u) = \prod_{v|p} L_{K,v}(\mathbf{W}_{n,K}, u).$$

If  $p$  splits in  $K$ , both residue fields equal  $\mathbf{F}_p$ . Because  $\mathbf{W}_{n,K}$  is the base change of  $\mathbf{W}_n$ , the two local factors are identical:

$$L_{K,p}(\mathbf{W}_{n,K}, u) = L_{\mathbf{Q},p}(\mathbf{W}_n, u)^2. \quad (10)$$

Apply the power-series logarithm normalized by  $\text{Log}_0(1) = 0$ . Multiplication by the C51 exponent  $2/n$  yields

$$\frac{2}{n} \text{Log}_0 L_{K,p} = \frac{4}{n} \text{Log}_0 L_{\mathbf{Q},p}.$$

The remaining denominator is  $n/\gcd(n, 4)$ . Hence the third row retains exponent  $4/3$ , while the fourth row has exponent one. Exponentiating in the same origin-normalized branch proves (7).

*Remark 4.1* (Meaning of ordinary). The fourth-row result is ordinary in rank and local multiplicity: one rank-255 rational factor occurs with exponent one. The packet contains Tate-normalized pieces, so its polynomial is naturally in  $\mathbf{Q}[T]$  and need not be integral. The integral statement in Theorem 2.4 concerns the untwisted rank-10 smooth-projective summand.

Theorem 2.2 now follows. For  $n \geq 5$ , the same denominator arithmetic is only conditional on smoothness and an analogous source-packet extraction. The all- $n$  theorem proved in this paper remains the equation descent.

## 5 The twisted dihedral group and Chow descent

The fourth  $K$ -model carries the source group

$$G = \text{Dih}(C_{12}) = \langle r, s \mid r^{12} = s^2 = 1, \, srs = r^{-1} \rangle, \quad |G| = 24.$$

Transport through the semilinear descent datum gives

$$\delta(r) = r^{-1}, \quad \delta(s) = sr^{-1} = rs. \quad (11)$$

Consequently

$$\delta(r^k) = r^{-k}, \quad \delta(r^k s) = r^{1-k} s. \quad (12)$$

These formulas define a Galois action by group automorphisms. Descent of the constant group  $G_K$  along this action gives a finite étale  $\mathbf{Q}$ -group scheme  $\mathcal{G}$  of rank 24, split by  $K$ . It is not constant: only 1 and  $r^6$  are fixed geometric elements. Thus the statement is not that twenty-four separate automorphisms are defined over  $\mathbf{Q}$ .

### 5.1 The invariant graph sum

On  $X_{4,K} \times X_{4,K}$ , let

$$e_G = \frac{1}{24} \sum_{g \in G} [\Gamma_g]. \quad (13)$$

Equation (12) permutes all twenty-four graph cycles, so  $\tau(e_G) = e_G$ . Let

$$q : (X_{4,0} \times X_{4,0})_K \longrightarrow X_{4,0} \times X_{4,0}$$

be quadratic base change and define

$$e_{\mathcal{G}} = \frac{1}{2} q_* e_G. \quad (14)$$

The pull-push identities for cycles [3, §§1.4, 1.7] give

$$q^* e_{\mathcal{G}} = \frac{1}{2} (e_G + \tau e_G) = e_G, \quad q_* q^* = 2.$$

Therefore  $q^*$  is injective on Chow groups with rational coefficients. The two denominators have different origins:  $1/24$  is Reynolds averaging, whereas  $1/2$  is field transfer.

Over  $K$ , the group average is an idempotent, is fixed by transposition, and commutes with  $\pi_5$ . Each assertion is a polynomial identity in correspondences. Pullback injectivity descends all of them to  $\mathbf{Q}$ . Define

$$\pi_{\text{core},0} = \pi_5 e_{\mathcal{G}}, \quad \pi_{\text{lev},0} = \pi_5 - \pi_5 e_{\mathcal{G}}. \quad (15)$$

These are mutually orthogonal self-transpose Chow projectors. This proves the descent assertion of Theorem 2.3. Rational coefficients are essential; the transfer argument does not supply an integral Chow projector.

## 5.2 Ranks and Hodge types

The frozen character calculation over  $K$  gives rank 10 for  $\pi_5 e_G$  and rank 158 for its middle complement. Since base change preserves realization ranks, (15) has the same ranks over  $\mathbf{Q}$ . The even packet has rank 87, so

$$\text{rank } W_4 = 87 + 10 + 158 = 255.$$

The raw core has Hodge realization

$$(4, 1)^1 + (3, 2)^4 + (2, 3)^4 + (1, 4)^1.$$

After one Tate twist, this becomes

$$(3, 0)^1 + (2, 1)^4 + (1, 2)^4 + (0, 3)^1,$$

which is of Calabi–Yau–threefold Hodge type. The source-normalized odd packet uses the second twist:

$$M_0(2) : (2, -1)^1 + (1, 0)^4 + (0, 1)^4 + (-1, 2)^1.$$

Neither Hodge ledger identifies an actual Calabi–Yau variety or proves that the motive is of abelian type.

## 6 Strictly compatible polynomials of the raw core

Let

$$M_0 = (X_{4,0}, \pi_{\text{core},0}, 0).$$

Spread  $X_{4,0}$ , the hyperplane class, and the rational correspondence  $\pi_{\text{core},0}$  over the complement of a finite set  $S$  of rational primes. At  $p \notin S$ , specialization of the projector commutes with geometric Frobenius  $F_p$ .

### 6.1 Rationality and coefficient-field independence

Choose an integer  $D > 0$  that clears the denominator of the projector. For every  $m \geq 1$ , the trace of  $\pi_{\text{core},0} F_p^m$  is a rational multiple of the trace of an integral algebraic correspondence. The correspondence comparison theorem of Katz–Messing makes these traces independent of the chosen Weil cohomology, hence independent of  $\ell$  [5, Theorem 2(2), pp. 76–77]. Applying Newton identities to the first ten traces gives one polynomial

$$\chi_{p,\text{core}}(U) = \det(U - F_p \mid H_\ell(M_0)) \in \mathbf{Q}[U] \quad (16)$$

of degree 10, independent of  $\ell$ .

No semisimplicity is used. The idempotent itself decomposes the realization into its image and kernel, both stable under Frobenius.



## 6.2 The separate integrality step

Denominator clearing in the cycle does not by itself prove integral coefficients. Instead, consider the full middle characteristic polynomial

$$\chi_{p,H^5}(U) = \det(U - F_p \mid H_{\text{ét}}^5) \in \mathbf{Z}[U].$$

The projector decomposition gives a factorization

$$\chi_{p,H^5}(U) = \chi_{p,\text{core}}(U)\chi_{p,\text{lev}}(U)$$

into monic polynomials in  $\mathbf{Q}[U]$ . Every root of either factor is a root of the full polynomial and is therefore an algebraic integer. The rational coefficients of each factor are algebraic integers, hence integers. Thus

$$\chi_{p,\text{core}}(U) \in \mathbf{Z}[U].$$

Reversing coefficients now gives

$$P_p(T) = \det(1 - F_p T \mid H_\ell(\mathbf{M}_0)) = T^{10} \chi_{p,\text{core}}(T^{-1}) \in \mathbf{Z}[T]. \quad (17)$$

The polynomial  $\chi_{p,\text{core}}$  is monic;  $P_p$ , whose constant term is 1, is not called monic.

## 6.3 Purity and reciprocity

The roots of  $\chi_{p,\text{core}}$  are eigenvalues on  $H^5$ , so they are pure of weight 5. The middle Poincaré pairing takes values in  $\mathbf{Q}_\ell(-5)$ , hence

$$\langle F_p x, F_p y \rangle = p^5 \langle x, y \rangle.$$

Self-transposition and idempotence give

$$\ker \pi_{\text{core},0} = (\text{im } \pi_{\text{core},0})^\perp,$$

so the restricted pairing is nondegenerate. The spread projector commutes with  $F_p$ , and its image is stable. Frobenius is therefore a similitude of multiplier  $p^5$  on the core, and its ten eigenvalues occur in pairs

$$\alpha, \quad \frac{p^5}{\alpha}.$$

Their product is  $p^{25}$ , and

$$P_p(T) = p^{25} T^{10} P_p\left(\frac{1}{p^5 T}\right). \quad (18)$$

Writing  $P_p(T) = \sum_{k=0}^{10} a_k T^k$  and comparing coefficients yields

$$a_{10-k} = p^{25-5k} a_k.$$

This argument uses the invariant direct summand supplied by the idempotent, not a semisimplicity hypothesis.

Finally, geometric Frobenius acts as  $p$  on  $\mathbf{Q}_\ell(-1)$ , hence as  $p^{-2}$  on  $\mathbf{Q}_\ell(2)$ . The normalized core  $\mathbf{M}_0(2)$  has weight 1 and polynomial

$$P_p^{(2)}(T) = P_p(T/p^2) \in \mathbf{Q}[T].$$

Its denominators are the prescribed Tate denominators, not a failure of compatibility. This completes Theorem 2.4.

## 7 Quadratic Artin formalism and proof boundaries

Let  $V$  be any of the rational realizations constructed above, and let  $\chi_K$  be the quadratic character of  $K/\mathbf{Q}$ . Induction and restriction give

$$\mathrm{Ind}_{G_K}^{G_{\mathbf{Q}}} \mathrm{Res}_{G_K}^{G_{\mathbf{Q}}} V \simeq V \oplus (V \otimes \chi_K).$$

The corresponding Artin formalism [1, §3, Proposition 3.8] gives, away from a common finite bad set,

$$L_K(V|_{G_K}, s) = L_{\mathbf{Q}}(V, s) L_{\mathbf{Q}}(V \otimes \chi_K, s). \quad (19)$$

Equation (19) is an identity of incomplete Euler products. We do not define bad-place Weil–Deligne factors here.

At a split prime,  $\chi_K(p) = 1$ , so the two local rational factors coincide. This is (10). At an inert prime, there is one  $K$ -place  $v$  with residue field  $\mathbf{F}_{p^2}$ , and

$$F_v = F_p^2.$$

If

$$P_p(U) = \prod_{i=1}^r (1 - \alpha_i U),$$

then

$$P_{K,v}(T) = \prod_{i=1}^r (1 - \alpha_i^2 T), \quad P_{K,v}(U^2) = P_p(U) P_p(-U). \quad (20)$$

The last expression is generally not a square. The split half-power is therefore a placewise identity, not the local shadow of a global square root.

### 7.1 One-prime irreducibility: exact scope

Suppose a future computation produces the full rank-10 polynomial  $P_p(T)$  at one good prime and proves it irreducible over  $\mathbf{Q}$ . Any Chow projector over  $\mathbf{Q}$  has a  $G_{\mathbf{Q}}$ -equivariant  $\ell$ -adic realization and hence commutes with  $F_p$ . If its image were nonzero and proper, the idempotent decomposition into image and kernel would factor  $P_p(T)$  over  $\mathbf{Q}$ . Irreducibility would therefore exclude every rational Chow projector whose cohomological image is nonzero and proper. Semisimplicity is not needed: the idempotent itself provides the direct sum.

This criterion does not exclude a projector with zero or full realization, which would require a faithfulness or nilpotence input. Nor does it exclude projectors after extending the coefficient field, prove absolute Chow indecomposability, or imply automorphy. The uncertified pre-C53 reconnaissance anchor records only the first raw trace  $-469$  at  $p = 7$ , not the full polynomial, so the criterion remains a future gate.

### 7.2 Geometric realization boundary

A conic-bundle or Prym realization is not part of the C53 certificate. A future route would first need an independent flatness theorem, followed by a discriminant and intersection-motive analysis. No such result is used in this paper.

## 8 Exact replay and adversarial gates

The accompanying certificate implements exact arithmetic in  $\mathbf{Q}[\rho]/(\rho^2 + \rho + 1)$ . It checks the family formulas for  $2 \leq n \leq 10$ , including the cocycle, fixed basis, determinant, and rational substitutions. These finite controls test the implementation; the symbolic argument in section 3 proves the all- $n$  theorem.

For the fourth row, the replay reconstructs all twenty-four group elements, verifies the transported group law on all  $24^2 = 576$  pairs, and partitions the elements into the two fixed points and eleven conjugate pairs. Separate gates enforce the Reynolds denominator 24, the field transfer denominator 2, and the distinction among the raw motive  $M_0$ , its Calabi–Yau-type twist  $M_0(1)$ , and the source-normalized twist  $M_0(2)$ .

The local-factor gates independently test the split exponent conversion and the inert identity (20). They reject a constant rational dihedral group, an inert half-root, a rank-255/2 compatible system, automatic integrality after a Tate twist, and any functional equation promotion.

At the release-candidate stage, the independent checker passes all 20 semantic gates, and the targeted mutation suite rejects all 63 alterations. The locked certificate, payload, independent-check, and code/results-manifest SHA-256 hashes are

```
f4325a5987933e2acf81656389d46701d82d38912c546d1e5996123f617f6e79,
8064224eda63fa9d890efd26ec9aa167c7cd9458662620be3135196a09494d41,
0d38643ded626c2a5e1536c8a4df9c56ae98c4fda01e1d15660996ea8c495e67,
b62f353d119d6c8565f513dad771a047a5e6343411d08ad2e91562fe84923480.
```

They form the final code/results evidence tuple. The full-project manifest and implementation-provenance backfill are separate release-integration steps.

The  $p = 7$  value is labeled `PRE_C53_RECONNAISSANCE_REGRESSION_ANCHOR_UNCERTIFIED`. It records raw core trace  $-469$ , or  $-67/7$  after the second Tate twist. The checker verifies the recorded arithmetic but does not independently reconstruct the fixed-locus counts, and no C52 provenance is asserted. This paper is not a table of Frobenius traces, and the anchor is not used to infer irreducibility.

## 9 Limitations and reproducibility

**Limitations.** The explicit equation descent holds for every  $n \geq 2$ , but the smooth motivic packet is certified only for  $n = 2, 3, 4$ . The exponent-one fourth-row factor is proved only at good split primes. The paper does not construct bad local factors, a global half-root, analytic continuation, a functional equation, automorphy, an actual Calabi–Yau threefold, or an all-correspondence indecomposability theorem.

**Reproducibility.** The certificate records the C52 implementation commit

```
208feef86365cd92ace8dad02904acff6623eeec
```

and the frozen C52 certificate hash

```
a2b0b281bfb311f979c7ed65e441a184ebe338b05f5fec8a60768610965c9c94.
```

The producer and independent checker use separate exact arithmetic paths. The default runner regenerates the certificate in a temporary directory, compares bytes, executes the mutation suite, and verifies the manifest. No fitted parameter, floating-point theorem gate, zero table, or post-hoc local projector is used.

**Data and ethics.** The work uses generated exact algebraic data and public mathematical literature; it involves no human subjects or personal data. The source chronology and every excluded promotion are machine-gated.

## A Additional proof details

### A.1 The fourth fixed matrix

In the ordered variables  $(u_0, u_1, v_1, u_2, v_2, u_3, v_3, u_4)$ , the coordinate change is

$$\begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{pmatrix} = \begin{pmatrix} u_0 \\ u_1 + \theta v_1 \\ u_2 + \theta v_2 \\ u_3 + \theta v_3 \\ (1 + \rho)u_4 \\ u_3 - \theta v_3 \\ \rho(u_2 - \theta v_2) \\ u_1 - \theta v_1 \end{pmatrix}.$$

The semilinear reversal is

$$M_4(x_0, \dots, x_7) = (x_0, x_7, \rho x_6, x_5, \rho x_4, x_3, \rho x_2, x_1).$$

Direct conjugation gives  $M_4\tau(B_4) = B_4$ , and

$$\det B_4 = (2\theta)^3 \rho(1 + \rho) = 24\theta.$$

### A.2 The quadratic edge ledger

For completeness, the transformed quadric has three edge classes:

$$\begin{array}{ll} i = 0 & : x_0 x_1 \mapsto x_0 x_{N-1} \quad \text{with phase 1,} \\ 1 \leq i \leq N-2 & : x_i x_{i+1} \mapsto x_{N-i} x_{N-i-1} \quad \text{with phase } \rho, \\ \text{closing} & : \rho x_{N-1} x_0 \mapsto \rho x_1 x_0. \end{array}$$

Thus the transformed closing coefficient is 1 and every nonclosing coefficient is  $\rho$ . Since  $\rho \cdot \rho^2 = 1$ , this is exactly  $\rho Q_{n, \rho^2}$ .

### A.3 The transported group automorphism

Let  $t = sr^{-1}$ . Then

$$t^2 = sr^{-1}sr^{-1} = 1, \quad tr^{-1}t = r.$$

Therefore  $r \mapsto r^{-1}$ ,  $s \mapsto t$  respects the dihedral presentation and defines an automorphism. It fixes  $r^k$  only when  $2k = 0 \pmod{12}$ , giving  $1, r^6$ , and fixes no reflection because  $2k = 1 \pmod{12}$  has no solution. Hence the Galois orbits have sizes

$$1, 1, \underbrace{2, \dots, 2}_{11 \text{ times}}.$$

The twist is outer. Indeed, an inner automorphism that inverts  $r$  is conjugation by a reflection, and sends  $s$  to  $r^{2a}s$  for some  $a$ ; it cannot send  $s$  to  $rs$ .

### A.4 Correspondence identities after transfer

Over  $K$ ,

$$e_G^2 = e_G, \quad {}^t e_G = e_G, \quad e_G \pi_5 = \pi_5 e_G.$$

The first formula follows by counting the 24 pairs  $(g, h)$  with  $gh = k$  for each  $k \in G$ ; the second uses closure under inversion; the third uses preservation of the hyperplane class. Pullback of  $e_{\mathcal{G}} = 2^{-1}q_*e_G$  is  $e_G$ . Since  $q^*$  is injective on rational Chow groups, all three identities hold over  $\mathbf{Q}$ . They imply

$$\pi_{\text{core},0}^2 = \pi_{\text{core},0}, \quad \pi_{\text{lev},0}^2 = \pi_{\text{lev},0}, \quad \pi_{\text{core},0}\pi_{\text{lev},0} = 0,$$

and self-transposition of both projectors.

## A.5 Four steps in reciprocity

The coefficient relation in Theorem 2.4 uses the following four facts.

1. The middle Poincaré pairing is

$$H^5 \times H^5 \longrightarrow \mathbf{Q}_\ell(-5).$$

With geometric Frobenius,  $\langle F_p x, F_p y \rangle = p^5 \langle x, y \rangle$ .

2. Since  $\pi_{\text{core},0}$  is a self-transpose idempotent,

$$\ker \pi_{\text{core},0} = (\text{im } \pi_{\text{core},0})^\perp.$$

The pairing restricted to its image is therefore nondegenerate.

3. The spread projector is defined over  $\mathbf{F}_p$ , so it commutes with  $F_p$ ; its image is Frobenius-stable.
4. A Frobenius similitude of multiplier  $p^5$  on this ten-dimensional nondegenerate summand has eigenvalues paired as  $\alpha, p^5/\alpha$ . Comparing coefficients in (18) gives

$$a_{10-k} = p^{25-5k} a_k.$$

No diagonalization or semisimplicity hypothesis enters these steps.

## A.6 Why one trace is not a polynomial

The uncertified pre-C53 reconnaissance anchor at  $p = 7$  records

$$\text{tr}(F_7 \mid \mathbf{M}_0) = -469, \quad a_1 = 469.$$

Reciprocity then determines  $a_9 = 7^{20} a_1$  and  $a_{10} = 7^{25}$ , but leaves  $a_2, a_3, a_4, a_5$  undetermined. The first trace therefore cannot certify irreducibility or a complete Euler factor.

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